

6.5 Hypothesis tests

Probability and Statistics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

State H_0 and H_1 with parameter and direction before calculating a test statistic. Compare p-value with significance level or statistic with critical region; final conclusion must be in context and must not claim H_0 proven. A one-sided test requires a justified directional claim made in advance.

Worked example: follow the reasoning

For $z=2.2$ in a right-tail test, $p \approx 0.0139$; reject at 5%, not at 1%.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A right-tail z test yields $p=0.032$. State decisions at 5% and 1% and explain what a 5% significance level controls. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Reject H_0 at 5% [1], fail to reject at 1% [1]; 5% is maximum long-run type I error probability under H_0 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.